

MAG TAPE SYSTEM (MAGTAP,24) 2/1/72 BPC

TGO, 0 /ENTRY IF BITS TO BE OR'ED IN
TGO+1, DAP 4CHAN3

LAW I 4
DAC TERCNT
TGOR, LAC I 4CHAN3
IOR UNIT
IOR TGO
LIA
IDX 4CHAN3
RRO 200
JSP THLT1
JMP I CH4BRK

TGO1, DZM TGO
JMP TGO+1

THLT1, LAW I 1
ADD 4CHAN3
DAC 4CHAN3
ISP TERCNT
JMP TGOR
JMP THLT

TERCNT, 0
↑L

/CHANNEL 4 TAPE SERVICE ROUTINE

```

4CHAN1:  LIO CONTR2
          SPI
          JMP I CH4BRK
          RIL 1S
          SPI I
          JMP I CH4BRK
          RRI 1200
          LAC I (DCH 72) /WASTE 15 MICRO-SECONDS
          RIR 8S
          SPI
          JMP I CH4BRK
          RRI 200
          SPI I
          IDX 4CHAN3
          JMP I 4CHAN3

```

```

4CHAN3.  DCH I FTAPE      /ALWAYS IN CORE 15

```

/TAPE WAIT ROUTINE

```

7WAIT.   DAC 7WAITX
          LIO UNIT
          LAW 0UNIT
          SNI I
          LAW 1UNIT
          DAP 7WAIT1
          LAC I CONTR2
          XOR I 7WAITX
          AND (7777
          XOR I 7WAITX
7WAIT1.  DAC .
          IDX 7WAIT1
          IDX CONTR2
          LAC I CONTR2
          XCT 7WAIT1
          IDX 7WAIT1
          IDX CONTR2
          LAC I CONTR2
          XCT 7WAIT1
          LAC 0CTR
          AAI
          DAC 7WAITX
          LAW I 1
          DAC I 7WAITX
          IDX CONTR2
          JMP FTAPES5

```

/UNIT STILL IN IO

```

7WAITX.  0
1SECK.   0
0UNIT.   REPEAT 3.      0
1UNIT.   REPEAT 3.      0
↑L

```

/CHANNEL 7 TAPE SERVICE ROUTINE

```
1SECJA.  DAC 1SECK
          LIO END2
          LAC 0UNIT
          DIO 0UNIT      /USE 0UNIT AS TEMP STORAGE
          DAC I 0UNIT
          IDX 0UNIT
          LAC 0UNIT+1
          DAC I 0UNIT
          IDX 0UNIT
          LAC 0UNIT+2
          DAC I 0UNIT
          IDX 0UNIT
          JMP 1SECJC
```

```
1SECJB.  DAC 1SECK
          LIO END2
          LAC 1UNIT
          DIO 1UNIT
          DAC I 1UNIT
          IDX 1UNIT
          LAC 1UNIT+1
          DAC I 1UNIT
          IDX 1UNIT
          LAC 1UNIT+2
          DAC I 1UNIT
          IDX 1UNIT
```

```
1SECJC.  SAD (TTBL+6
          LAC (TTBL
          LSM
          DAC END2
          LAC CONTR2
          IOR (200000
          DAC CONTR2
          ESM
          ISB 400
          JMP I 1SECK
```

↑L

/TAPE DISPATCH ROUTINE

BFTAPE:

```
FTAPE,      NOP
             LAC CONTR2
FTAPE+2,     AND (177777)
             SAD END2
             JMP FTAPE6
             DAC FTAPE1
             CLA
             LIO I FTAPE1
             RCL 6S
             ADD (FTAPET
             DAP FTAPE2
             CLA
             RCL 1S
             DAC UNIT
             SAL 1S
             ADD (UNTBL
             DAC UNPTR
             CLA
             RCL 2S
             RAR 8S
             DAC TAPSTT
             RIR 9S
             DIO I TUSER
             IDX FTAPE1
             LAC I FTAPE1
             DAC TAPPAR
             IDX FTAPE1
             DZM ERWRD2
             CLI
             LAC I FTAPE1
             RCL 6S
             LFI
             LAC TAPPAR
FTAPE2.     XCT .
```

↑L

/USER WANTS DATA CHANNEL

```
TPRLS1.  LAC I UNPTR    /MARK AS NO LONGER AT LOAD POINT
          RAL 2S         /ARE WE IN FORWARD POSITION
          SMA
          JMP TPRLS2     /IN FORWARD POSITION
          JSP TGO1
          204000
          JSP THLT
TPRLS4.   JSP 7WAIT
          040000

TPRLS3.   JSP TGO1
          200000
          JMP TPRLS4
          RRI 1200
          LAC I (DCH 72) /WASTE 15 MICROSECONDS
          RIL 2S
          SPI
          JMP TPRLS4
          LAC I UNPTR    /MARK IN FORWARD POSITION
          XOR (100000
          DAC I UNPTR

TPRLS2.   LAC (550000    /SET TO USE CHANNEL
          LSM
          DIP CONTR2
          CLA            /IS CHANNEL BUSY
          SAD I DCONT
          ISB 100
          ESM
FTAPE3.   LAW FTAPE
          DAP 4CHAN3
          JMP I CH4BRK

FTAPE6.   DAC CONTR2
          JMP FTAPE3

FTAPET.   NOP
          JMP 4SPC
          JMP 4RWD
          JMP 4ENT
          JMP TPRLS3
          JMP LDPT
          JMP LDPT1
          JMP 4WEOF
```

↑L

/RELEASE USER ON CHANNEL 4

```
TPRLS,      LAC I TUSER
             SUB MSTAT
             DAP FTAPE4
             LAC (377777
             AND I FTAPE4
             DAC I FTAPE4
             ISB 1700
             DZM I HOTFLG
             LAW 3
             ADD CONTR2
FTAPES5.    AND (177777
             SAD (TTBL+6
             LAC (TTBL
             IOR (200000
             DAC CONTR2
             JMP FTAPE+2
```

```
FTAPE4.     DCH .
FTAPE1.     0
UNIT.       0
TAPPAR.     0
UNPTR.      0
TAPSTT.     0
```

/CHANNEL 16 TAPE INITIALIZE ROUTINE
/LINK SET-RETURN 3

```
16TAPE.     DAP 16TAPX
             JSP I .SETUP
             LAC I (DCH 72)
             LIA
             AND (-1)
             SZA
             JMP NOTYOURS
             JSP 16TAP2
             AND (200177
             SAS I 16USER
             JMP NOTYOURS
             LIO I 16TAP1
             LAC I (DCH 70)
16TAPX.     JMP .
```

```
16TAP1.     0
16UNIT.     0
+L
```

/TAPE WAITING LIST ROUTINE

```

TAPE.      0
            DAC TAPE3          /SAVE POINTER TO COMMAND
            JSP I RELEASE
            LAC I TAPE3
            IOR 16UNIT
            IOR I 16USER
            DSC 700            /DE-ACTIVATE CHANNEL 7 SO END2 WILL NOT CHANGE
            LIO END2           /GET END OF RING BUFFER
            DIO TAPE1
            DAC I TAPE1        /SAVE FIRST WORD
            IDX TAPE1
            LAC TAPE
            DAC I TAPE1
            IDX TAPE1
            LIF
            RCR 6S
            DIP TAPE3
            IDX TAPE3          /GET RETURN ADDRESS
            DAC I TAPE1
            IDX TAPE1
            SAD (TTBL+6)
            LAC (TTBL)
            LSM
            DAC END2
            LAC CONTR2
            IOR (200000)
            DAC CONTR2
            ASC 700
            FSM
            ISB 400
            JMP I R0

```

```

TAPE1,      0
TAPE3,      0
CONTR2.     TTBL
END2.       TTBL

```

```

16TAP2.     DAP 16TP2X
            DIO GTAPE7
            RIR 7S
            LAI
            SZF 4
            IOR (2000)
            SZF 5
            IOR (1000)
            DAC 16UNIT
            RIL 8S
            LAI
            ADD (UNTBL)
            DAC 16TAP1
            LAC I 16TAP1
16TP2X.     JMP .

```

+L

/REWIND ICT

/FLAG 1=1-WITH INTERLOCK, 0-WITHOUT INTERLOCKTL

RWD: JSP 16TAPE
JDA TAPE
020000

4RWD. SZF 1
JMP RWD2 /WITH INTERLOCK
JSP TGO1
200000
JSP THLT1
RRI 1200
LAC I (DCH 72) /WASTE 15 MICROSECONDS
RIL 8S

SPI
JMP LDPT4
JSP TGO1
210000
JSP THLT
LDPT3, JSP 7WAIT
050000 /WAIT FOR LOAD POINT

RWD2. JSP TGO1
212000
JSP THLT1
LAC (200000
DAC I UNPTR
IDX UNPTR
CLC
DAC I UNPTR
JMP TPRLS

↑L

/SPACE TAPE

/FLAG 1=1←BACKWARDS, 0←FORWARDS

/FLAG 2=1←FILES, 0←BLOCKS (FILES BACKWARDS IS ILLEGAL)

SPC: JSP 16TAPE
SPC1. SPA
JMP 16ILLSPC
SZA I
JMP I R0
JDA TAPE
010000

4SPC. SZF I 2
JMP SNB /SPACE BLOCKS
DAC ERWRD2
CMA
DAC TAPPAR

SNF4. CLF 3
SNF1. LAW 7777
JDA .SFNB
JMP SNF1
SZF I 3 /EOF-WAS IT FIRST
JMP SNF3
LAC .SFNB
SAD (7776
JMP SNF5
ISP TAPPAR
JMP SNF1
JMP TPRLS

SNF5. LAC ERWRD2
ADD TAPPAR
JMP SNF6

/FIRST END OF FILE
SNF3. STF 3
ISP TAPPAR
JMP SNF1
JMP TPRLS

SNB. JDA .SFNB
JMP TPRLS /GOOD RETURN
LAW 1

ADD .SFNB
SNF6. DAC ERWRD2
4EOF. LAW 2000
JDA TAPERR

↑L

/SUBROUTINE TO SPACE FORWARD N BLOCKS

```

.SFNB.      0
            DAP .SFNBX
            LAC (260000)
            SZF I 1
            JMP .SFNB5
            LAC I UNPTR
            IOR (1000000)
            DAC I UNPTR
            LAC (270000)
.SFNB5.     DIP .SFNB3+2      /SET WHETHER FORWARD OR BACKWARDS
.SFNB1.     LAC .SFNB
.SFNB1+1.   SPO
.SFNBX.     JMP .            /GOOD RETURN
            SUB (77)
            DAC .SFNB
            SPA
            JMP .SFNB2      /LAST TIME
            LAW 1760
.SFNB3.     IOR TAPSTT
            JDA TGO
.SFNB3+2.   0
            JMP .SFNB4
            JMP .SFNB1
.SFNB4.     RRI 200
            LAI
            RRI 1200
            AND (77)        /RESET BLOCK COUNT
            ADD .SFNB
            DAC .SFNB
            RIL 8S          /HIT LOADPOINT?
            SPI
            JMP 4LDPT
            RIL 1S          /PERHAPS AN EOT
            SPI
            JMP 4EOT
            RRI 200         /CHECK FOR DATA ERROR
            LAI
            RAL 2S
            SPA
            JMP THLT        /UNRECOVERABLE
            AND (70000)
            SZA
            JMP .SFNB1      /RE-EXECUTE FOR DATA ERRORS
            IDX .SFNBX
            JMP .SFNBX
4EOT.      LAW 1000
            JDA TAPERR
.SFNB2.     ADD (77)
            SAL 4S
            DZM .SFNB
            JMP .SFNB3

```

↑L

/WRITE END OF FILE

WEOF: JSP 16TAPE

RIL 3S

/TAPE STATUS IN IO

SPI I

JMP WRING

JDA TAPE

070000

4WEOF, CLF 2

LAW I 4

DAC TAPPAR

/ERROR COUNT

WEF2.

LAC TAPSTT

AND (-2000)

JDA TGO

230000

JMP WEFERR

JMP TPRLS

WEFERR.

RRI 200

LAI

AND (16000)

SZA

JMP WEFER1

/DATA ERROR

RRI 1200

LAC I (DCH 72) /WASTE 15 MICROSECONDS

RIL 9S

SPI

JMP 4EOT

RIL 2S

SPI I

JSP THLT

WEFER1.

LAC TAPSTT

JDA TGO

270000

JSP THLT1

ISP TAPPAR

JMP WEF2

SZF 2

JSP THLT

/DATA ERROR

JSP TGO1

216000

JSP THLT

STF 2

LAW I 4

DAC TAPPAR

JMP WEF2

↑L

/ERASE TAPE

ERASE: JSP 16TAPE
RIL 3S
SPI I
JMP WRING
SZA I
JMP I R0
SPA
JMP 16ILLSPC
JDA TAPE
030000

4ENT.

CMA
DAC TAPPAR

4ENT1.

JSP TGO1
216000

JSP THLT1

/NO ERRORS WE CAN DO ANYTHING ABOUT

RRI 1200

/EXCEPT EOT (WHICH ISN'T AN ERROR)

LAC I (DCH 72)

/KILL SOME TIME (15 MICROSECONDS)

RIL 9S

SPI

JMP 4EOT

ISP TAPPAR

/COUNT 5 INCHES

JMP 4ENT1

/AND GO BACK FOR MORE

JMP TPRLS

+L

TPRT. DAP TPRTX /TYPE OUT FOR MAGNETIC TAPE
 LIO I 16USER /TYPE OUT PROG. NO.,
 RIL 9S
 LAW I 3
 LSM
 JDA SOTOCT
 LIO I (BILLTT) /TYPE OUT TT.
 RIR 6S
 LAW I 2
 JDA SOTOCT
 LAC (TTYP1)
 JDA SOTXT /TYPE FIRST PART OF MESSAGE
 LAC I (DCH 72)
 SPA
 CMA
 JDA SOTPRT
 LAC (TTYP2)
 JDA SOTXT /TYPE SECOND PART OF MESSAGE
 LIO GTAPE7
 RIR 3S
 LAW I 1
 JDA SOTOCT /TYPE UNIT NO.
 LIO I (DCH 72)
 LAC (TTYP3)
 SPI
 JDA SOTXT
 CLC /TYPE CARRIAGE RETURN
 JDA SOTDCH
 ESM
 JMP .

TPRTX.
+L

```

/TAPE CONTROL IOT'S
TPCNT:      SZF 1          /GET OR RELEASE TAPE
            JMP RTAPE
GTAPE,      JSP I .SETUP
            CLI
            LAC I (DCH 72) /SEE IF REEL WANTED IS ALREADY ON TAPE DRIVE
            SPA
            CMA
            SAD .UNTBL+1   /UNIT 0
            JMP GTAPE1     /YES-GO SEE IF AVAILABLE
            LIO (1
            SAD .UNTBL+3   /UNIT 1
            JMP GTAPE1     /YES-GO SEE IF AVAILABLE
            CLA           /NOT ON TAPE DRIVE-IS THERE A USEABLE DRIVE
            DZM GTAPE7
            LIO .UNTBL     /CHECK STATUS OF DRIVE 0
            RCL 25
            SAD (1        /IS IT USEABLE AND FREE
            JMP GTAPE4     /YES
            IDX GTAPE7     /LOOK AT DRIVE 1
            LIO .UNTBL+2
            CLA
            RCL 25
            SAS (1        /IS UNIT 1 USEABLE AND FREE
            JMP EMPTY     /NO DRIVES AVAILABLE

```

```

GTAPE4.     LIO GTAPE7
            JSP 16TAP2
            JSP TPRT      /TYPE OUT ON SOROBAN
            JSP GSET
            IDX 16TAP1
            LAC GTAPE7
            LIO I (DCH 72)
            DAC I (DCH 72) /PUT UNIT IN IO
            LAC I 16TAP1
            SPI
            CMI
            DIO I 16TAP1   /SAVE REEL NUMBER
            SAS (-3)
            JMP GTAPE9
            LAC I (DCH 70)
            JDA TAPE
            050000

```

```

GTAP14.     SIR 15
            DIO I (DCH 72)
            SZF I 3
            JMP I R0
            LAC I (DCH 70)
            JMP SPC1

```

```

GTAPE7.     0
            +L

```

```

LDPT1,   JSP TGO1      /REWIND WITH INTERLOCK
          212000
          JSP THLT1     /ERROR IS UNFORGIVEABLE
LDPT.    JSP TGO1      /NOP TO CONNECT TO CHANNEL
B1,      200000
          JMP LDPT3     /NOT ACCEPTING COMMANDS YET
LDPT2,   LAC .         /MARK AS AT LOAD POINT AND WRITE RING IN
          AND I UNPTR   /ALSO MARK FORWARD POSITION
          IOR .
          DIP I UNPTR
          RRI 1300      /TAPE FAULT DIFFERENT IF WRITE RING OUT
          LAC I (DCH 72) /WASTE 15 MICROSECONDS
          LAI
          RRI 1200
          AND (-0
          XAI
          RAL 1S
          SMA
          JMP LDPT4
          LAC (40000)   /REMOVE BIT
          XOR I UNPTR
          DIP I UNPTR
LDPT4,   SZF I 3
          JMP TPRLS
          LAC TAPPAR    /IS IT LEGAL COUNT
          SPA
          JMP 4ILLSPC
          SZA I         /SPACE 0?
          JMP TPRLS     /YES, DONE
          JMP 4SPC

/TAPE ON DRIVE-SEE IF AVAILABLE
GTAPE1,  JSP 16TAP2
          XOR I 16USER  /CHECK IF ALREADY OWNED
          AND (7777)
          SZA I
          JMP GTAP15
          LAC I 16TAP1  /NOT ALREADY YOURS IS IT FREE
          RAL 1S
          SMA
          JMP NOTYOURS  /DRIVE ALREADY TAKEN-CAN NOT HAVE
          JSP GSET
GTAPE15. LAC I 16TAP1  /YOURS: DO YOU WANT TO CHANGE
          RAL 3S
          XOR I (DCH 72)
          SMA
          JMP GTAP14    /NO: RETURN TO USER
          JSP TPRT      /TYPE OUT ON SOROBAN
          LIO GTAPE7
          DIO I (DCH 72)
GTAPE9.  LAC I (DCH 70)
          JDA TAPE
          060000      /WAIT FOR LOAD POINT

```

+L

/RELEASE TAPE

RTAPE. JSP 16TAPE /GO CHECK IF YOUR UNIT ETC.

LAW I 7777

NAI

IOR (200000)

/MARK AS NOT HELD

DAC I 16TAP1

CLF 1

JSP TAPE+1

20000

/REWIND TO LOAD POINT

GSET.

DAP GSETX

LAC I 16USER

XOR I 16TAP1

AND (207777)

XOR I 16TAP1

DAC I 16TAP1

GSETX.

JMP .

TTYP1: 444523

005600

TTYP2: 004645

005600

TTYP3: 005156

↑L

/READ PHYSICAL RECORD
 /AC=CORE LOCATION
 /IO=UNIT NO.
 /WORD AFTER IOT IS LENGTH OF RECORD

```

RPR:      SZF 3
          STF 1
          SZF I 1
          CLL+UCML      /MARK AS RETURN THREE IOT
          JSP 16TAPE
          SZF I 2      /IS IT A WRITE
          JMP RPR6      /NO
          RIL 3S      /YES-IS WRITE RING IN
          SPI I
          JMP WRING
RPR6.     LAW 1
          ADD I RPC      /GET POINTER TO RECORD LENGTH
          DAC TAPE      /UUSE AS TEMP STORAGE
          LAW 28.      /CHECK IF BCD : LENGTH 28. IF IMPLIED
          SZF 4
          LAW 200.+10
          SZF I 1
          LAC I TAPE      /LENGTH NOT IMPLIED
          AND (7777)      /SET TO TWELVE BITS
          JDA TAPE
          000000
          CLF 1      /RESET FLG 1 FOR IOPMAX CHECK
          LAC TAPPAR      /IS IT A ZERO LENGTH BLOCK
          SZA I
          JMP I SWAP      /DO NOTHING
          SZF 2
          JMP RPR5      /IF WRITE SKIP IOPMAX CHECK
          LAC I AC
          SUB BOUND      /CHECK FOR PROTECTION ERROR
          SPA
          JMP ILLSPC
          ADD BOUND
          ADD TAPPAR      /ADD IN LENGTH
          SUB (I)
          SZM
          CLA+USTF 1
          ADD (I)
          SUB I AC
  
```

+L

```
RPR5.      DAP WPRC4
RPR5+1.    LAW 7777
           AND I AC
           IOR I SCORE
           DAC WPRC3      /SET DAC COMMAND
RPR2.      LAC UNIT
           IOR TAPSTT
           IOR (240000)
           SZF 2
           XOR (600000)
           DAC WPRC2
           JSP I GO
           I WPRC1
           RRI 700
           LAC (JMP)
           IAI
           SUB WPRC3
           DAC I ERCOD2
           SZF I 1
           JMP I SWAP
MAXERR.    LAC (100000)
           JMP 1TERR

WPRC1.     560002
WPRC2.     220000
WPRC3.     600000
WPRC4.     140000
           000000
```

```
START
↑L
```